AUO Display+

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Document Chinese Title**: 21.3" (G213UAN01.1) TFT-LCD 進料檢驗規範 (A grade)**

Document English Title**:** Incoming Inspection Standard For 21.3" (G213UAN01.1) TFT-LCD Modules (A)

# AU OPTRONICS CORPORATION

## Specification for Approval

## INCOMING INSPECTION STANDARD FOR 21.3" TFT-LCD MODULES ( A grade)

## Model Name: G213UAN01.1

| Approved By | Prepared By |
|---|---|
| YenChun Fu | YH Chuang |

**General Display Business Unit/AU Optronics**

| Customer | Checked and Approved by |
|---|---|
| | |

| Inspection criteria's change record | | | | |
|---|---|---|---|---|
| **Version/ Effective Date (Approval)** | **Page** | **Inspection Item** | **Previous Inspection Spec** | **New Inspection Spec** |
| V1 / 2026.04.21 | All | All | | Initial Specification |
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# 1 Scope

1.1 The incoming inspection standards shall be applied to TFT-LCD Modules (hereinafter called "Modules") that are supplied by AUO Display Plus Corporation (hereinafter called "seller").

1.2 Specifications contains

- Electrical inspection specifications
- Appearance specifications

## 2 Incoming inspection

The buyer shall inspect the Modules within the inspection period (ninety calendar days starting from the date of delivery.) The results of the inspection (acceptance or rejection) shall be recorded in writing and a copy will be promptly sent to the seller.

The buyer may, under reasonable commercial rejection procedures~~,~~ reject an entire lot in the delivery involved. Within the inspection period,if the Modules in the lot show a number of unacceptable defects in accordance with this Incoming Inspection Standards, the buyer must notify the seller in writing of any such rejection promptly, within three business days during the inspection period.

If the buyer fails to notify the seller within the inspection period, the buyer's right to reject the Modules shall be lapsed and the Modules shall be deemed accepted by the buyer.

If any questions arise in regards to the judgement, the outcome shall be decided upon the consultation between both parties.

The buyer shall sign the IIS back to the seller once the buyer verifies the model completely and moves to the mass production stage. The seller has the right to follow internal specifications or ES (engineeering sample) performances if the buyer fails to sign back to the seller.

## 3 Inspection sampling method

Unless otherwise agreed in writing, the method of incoming inspection shall be based on ANSI/ASQL Z1.4-2003.

3.1 Lot size: Quantity per shipment lot per model.

3.2 Sampling type: Normal inspection, single sampling.

3.3 Sampling level: Level II.

3.4 Acceptable quality level (AQL):

Major defect: AQL=1.0%.

Minor defect: AQL=2.5%.

| Lot Size | General Inspection Levels | | | Specific Inspection Levels | | | |
|---|---|---|---|---|---|---|---|
| | I | II | III | S-1 | S-2 | S-3 | S-4 |
| 2-8 | A | A | B | A | A | A | A |
| 9-15 | A | B | C | A | A | A | A |
| 16-25 | B | C | D | A | A | B | B |
| 26-50 | C | D | E | A | B | B | C |
| 51-90 | C | E | F | B | B | C | C |
| 91-150 | D | F | G | B | B | C | D |
| 151-280 | E | G | H | B | C | D | E |
| 281-500 | F | H | J | B | C | D | E |
| 501-1200 | G | J | K | C | C | E | F |
| 1201-3200 | H | K | L | C | D | E | G |
| 3201-10000 | J | L | M | C | D | F | G |
| 10001-35000 | K | M | N | C | D | F | H |
| 35001-150000 | L | N | P | D | E | G | J |
| 150001-500000 | M | P | Q | D | E | G | J |
| 500000 and over | N | Q | R | D | E | H | K |

| Sample size code letter | Sample size | Acceptance Quality Limits, *AQLs*, in Percent Nonconforming Items and Nonconformities per 100 Items (Normal Inspection) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0.010 | | 0.015 | | 0.025 | | 0.040 | | 0.065 | | 0.10 | | 0.15 | | 0.25 | | 0.40 | | 0.65 | | 1.0 | | 1.5 | | 2.5 | | 4.0 | | 6.5 | | 10 | | 15 | | 25 | | 40 | | 65 | | 100 | | 150 | | 250 | | 400 | | 650 | | 1000 | |
| | | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re | Ac | Re |
| A | 2 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↓ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | 30 | 31 |
| B | 3 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | 30 | 31 | 44 | 45 |
| C | 5 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | 30 | 31 | 44 | 45 | ↑ | |
| D | 8 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | 30 | 31 | 44 | 45 | ↑ | | ↑ | |
| E | 13 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | 30 | 31 | 44 | 45 | ↑ | | ↑ | | ↑ | |
| F | 20 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| G | 32 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| H | 50 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| J | 80 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| K | 125 | ↓ | | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| L | 200 | ↓ | | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| M | 315 | ↓ | | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| N | 500 | ↓ | | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| P | 800 | ↓ | | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| Q | 1250 | 0 | 1 | ↑ | | ↓ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |
| R | 2000 | ↑ | | ↑ | | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 6 | 7 | 8 | 10 | 11 | 14 | 15 | 21 | 22 | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | | ↑ | |

↓ = Use the first sampling plan below the arrow. If sample size equals, or exceeds, lot size, carry out 100 percent inspection.
↑ = Use the first sampling plan above the arrow.
Ac = Acceptance number.
Re = Rejection number.

# 4 Inspection instruments

4.1 Pattern generator: LD-2000 or equivalent model.

4.2 Video board: ADP video board or equivalent. The output of the signal should comply with the specification provided by ADP.

4.3 Luminance colorimeter: Topcon BM-7 or equivalent model

# 5 Inspection environment conditions

5.1 Room temperature : **20 ~ 25** °C.

5.2 Humidity: **65±5**% RH.

5.3 Illumination: Fluorescent light (Day-Light Type) display surface illumination to be 300 ~ 700 Lux. (standard 500Lux.)

5.4 Distance: **35±5** cm in front of LCD unit with a viewing angle of θ =10°.(refer to figure 1)

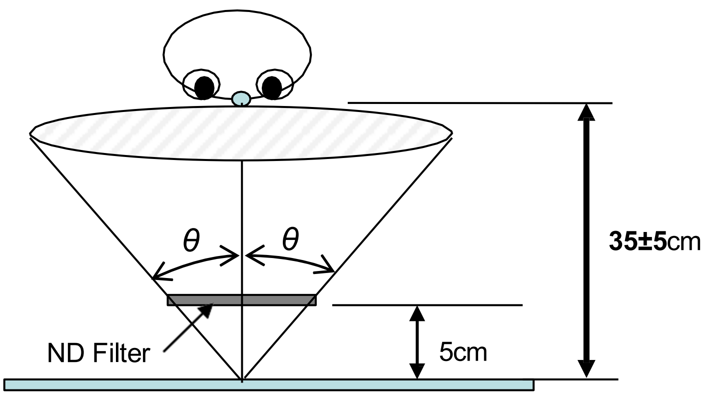



Fig. 1

5.5 Take off the protector of the polarizer while inspecting the display area.

5.6 If there is any question with the inspection, check the panel again when powered on.

# 6 Classification of defects

Defects are classified as major defects and minor defects according to the degree of defectiveness defined herein.

**Major defects:**

A major defect is a defect that is likely to result in failure, or to reduce materially the usability of the product for its intended purpose.

**Minor defects:**

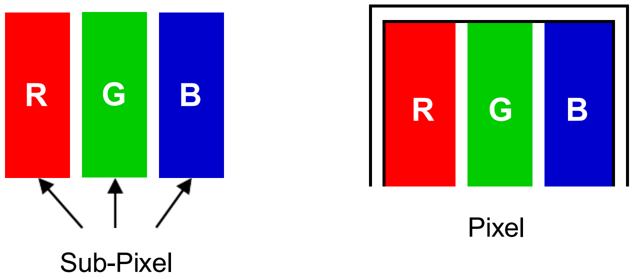
A minor defect is either a defect that is not likely to reduce materially the usability of the product for its intended purpose, or a deviation from an intended purpose with little bearing on the effective use or operation of the product.

6.1 Electrical inspection specification

| No. | Inspection Item | | Max. acceptable number / Distance | Unit | Note |
|---|---|---|---|---|---|
| 1 | Bright dots | 1 dot | 1 | dot | 1, 2, 6 |
| 2 | | 2 Adjacent dots | 0 | pair | 4, 5, 6 |
| 3 | | 3 Adjacent dots | 0 | pair | 4, 5, 6 |
| 4 | | Over 3 Adjacent dots | 0 | pair | 4, 5, 6 |
| 5 | | Distance between bright dots | - | mm | 3 |
| 6 | | Total (1+2+3+4) | 1 | dot | |
| 7 | Dark dots | 1 dot | 6 | dot | 1, 6 |
| 8 | | 2 Adjacent dots | 2 | pair | 4, 5, 6 |
| 9 | | 3 Adjacent dots | 0 | pair | 4, 5, 6 |
| 10 | | Over 3 Adjacent dots | 0 | pair | 4, 5, 6 |
| 11 | | Distance between dark dots | $\geq$ 20 | mm | 3 |
| 12 | | Total (7+8+9+10) | 6 | dot | |
| 13 | Bright dot and Dark dot | 2 Adjacent dots | 0 | pair | 4, 5, 6 |
| 14 | | Distance between dots | - | mm | 3 |
| 15 | Total amount of Dot Defects (1~13) | | 6 | dot | |
| 16 | Line defect | | Not allowed | - | |
| 17 | Mura | | Use 5% ND filter or judged by equivalent limit sample | | 6, 7, 8 |

- Note 1 ) Definition of pixel and sub-pixel: refer to figure 2.

Fig. 2




- Note 2 ) Bright dot and Small bright dot judgment criteria.

| Judgment Criteria | | Bright Dot Defect area | |
|---|---|---|---|
| | | ＞1/2 sub-pixel | ≦1/2 sub-pixel |
| ND filter 10% | Visible | Bright Dot | Small Bright Dot |
| | Invisible | Small Bright Dot | Small Bright Dot |

The drawing of 1/2 area sub-pixel definition: The 1/2 area sub-pixel can be defined as below one or more of specific shapes (Fig. 3). The amount of small bright dots should be less or equal to 10.

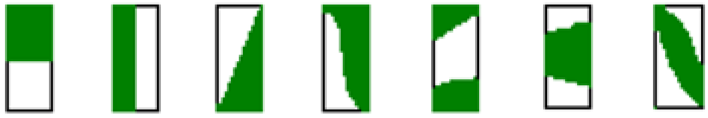
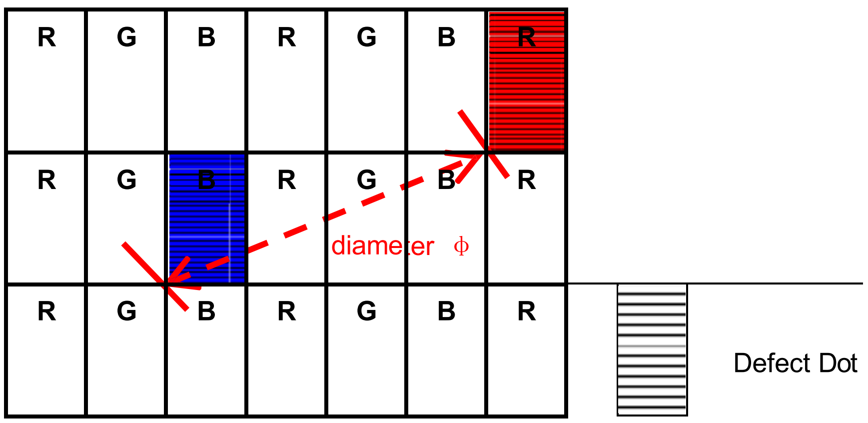


Fig. 3

- Note 3 ) Definition of diameter φ as following:

Fig. 4




- Note 4 ) Adjacent-dot defect should be observed under the same display pattern in any one of Black/Green/Blue/Red pattern.

Adjacent-dot defect : refer to Figure 5, dot 1,2,..,8 around A are all A's adjacent dots

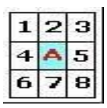



Fig. 5

- Note 5 ) In three (or more) adjacent dot defect, for any 3rd dot that adjacent to 2 continuous defective dots (can be of any combination of bright dots and dark dots), the 3rd dot, no matter how large it may be, should be viewed as a dot.
- Note 6 ) Inspection pattern: Standard inspection patterns of dot defect are listed below. ADP uses these patterns as standard criteria for judging dot defect. Please inform ADP if any other pattern is to be used to examine it.

| Test Pattern | Defect |
|---|---|
| Black | For bright dot(s) |
| Red | For bright and dark dot(s) |
| Green | For bright and dark dot(s) |
| Blue | For bright and dark dot(s) |

- Note 7 ) Unless specified in other written document or limit samples, Mura (display non-uniformity) should be inspected under the ND filter and be acceptable when it is invisible through ND filter.

  ND filter use method: The inspection method of ND Filter - holding ND filter in front of the panel around 5 cm and examine the panel from **35±5** cm in the front view for **3** seconds. (Fig. 1)
- Note 8 ) While operating over 50°C ambient temperature, there should be no function failure occur and Mura (display un-uniformity ) should be invisible under 1 % ND filter applied.
- Note 9 ) Image Retention : 5 seconds test pattern and image retention must be disappeared in 5 seconds after pattern changed.

6.2 Scratches, dent, extraneous substances and Appearance inspection specification

<table>
<tr><th rowspan="2">Judge area</th><th rowspan="2" colspan="2">Judge item</th><th rowspan="2" colspan="3">Inspection specification</th><th colspan="2">Judge criterion</th></tr>
<tr><th>Major</th><th>Minor</th></tr>
<tr><td rowspan="11">Active area</td><td rowspan="8">Particles on the polarizer</td><td rowspan="4">Circular</td><td colspan="2">Average diameter: D (mm)</td><td>Numbers</td><td rowspan="4"></td><td rowspan="4">○</td></tr>
<tr><td colspan="2">D≦0.3</td><td>Disregarded</td></tr>
<tr><td colspan="2">0.3 < D≦0.70</td><td>n≦7</td></tr>
<tr><td colspan="2">0.70 < D</td><td>n＝0</td></tr>
<tr><td rowspan="4">Linear</td><td colspan="2">Width: W (mm)<br>Length: L (mm)</td><td>Numbers</td><td rowspan="4"></td><td rowspan="4">○</td></tr>
<tr><td>W≦0.1</td><td>L≦ 5</td><td>Disregarded</td></tr>
<tr><td>0.1 < W≦0.3</td><td>5 < L≦7</td><td>n≦7</td></tr>
<tr><td>0.3 < W</td><td>7 < L</td><td>n＝0</td></tr>
<tr><td rowspan="3">Scratch/Dent on the polarizer</td><td rowspan="3">Circular</td><td colspan="2">Average diameter: D (mm)</td><td>Numbers</td><td rowspan="3"></td><td rowspan="3">○</td></tr>
<tr><td colspan="2">D≦0.3</td><td>Disregarded</td></tr>
<tr><td colspan="2">0.3 < D≦0.70</td><td>n≦7</td></tr>
</table>

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | 0.70 < D | n=0 | | |
| | | Linear | **Width: W (mm)**<br>**Length: L (mm)** | **Numbers** | | ○ |
| | | | W≦0.1 and L≦5 | Disregarded | | |
| | | | 0.1 < W or 5 < L | n=0 | | |
| | Bubble on the polarizer | Circular | **Average diameter: D (mm)** | **Numbers** | | ○ |
| | | | D≦0.3 | Disregarded | | |
| | | | 0.3 < D≦0.70 | n≦7 | | |
| | | | 0.70 < D | n=0 | | |
| Bezel | Front Bezel Gap | | ≦ 2.1mm | | | ○ |
| | Scratches, Warp, Sunken and Color Difference | | 1. No sharp burr/edge to cause safety issue.<br>2. The bezel defect to be accepted if it is not affecting display performance. | | | ○ |
| | Assembly Fail | | Not allowed | | ○ | |
| Label (S/N, B/L, WEEK) | No label | | Not allowed | | ○ | |
| | Upside Down | | | | ○ | |
| | Content Error | | | | ○ | |
| | Dirt | | Word can be read.<br>Barcode can be scanned. | | | ○ |
| | Not clear | | | | | ○ |
| | Font Deformation | | | | | ○ |
| | Label Broken | | | | | ○ |
| | Crease | | | | | ○ |
| | Label overlapping | | | | | ○ |
| | Position | | Be attached on right position | | | ○ |
| Screw | Missing screw | | Not allowed | | ○ | |
| | Loose | | Not allowed | | ○ | |
| Connector | Appearance | | As long as no function issues CN can be allowed. | | ○ | |

- Note 9 ) Extraneous substances which can be wiped out, such as fingerprint and particles are not considered as a defect.
- Note 10 ) Defects on the Black Matrix are not considered as a defect.
- Note 11 ) Defect size definition: (Unit:mm). When L≧2W, defect count as liner defect.

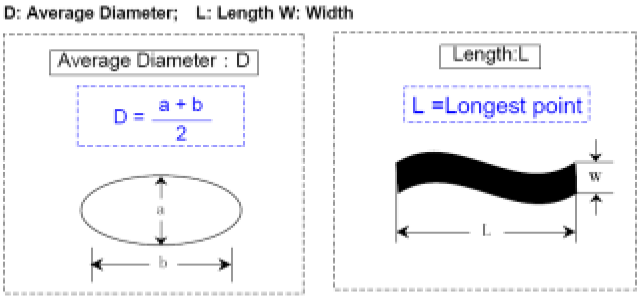



Fig. 6

# 7 Inspection judgement

7.1 The judgement of the shipped lot (acceptance or rejection) should follow the sampling plan of ANSI/ASQL Z1.4-2003, single sampling, normal inspection, level II.

7.2 If the number of defects is equal to or less than the applicable acceptance level, the lot shall be accepted.

7.3 If the number of defects is more than the applicable acceptance level, the lot shall be rejected and the buyer should inform the seller of the result of incoming inspection in writing.

# 8 Precaution:

Please pay attention to the following items when you use the LCD Module with back-light unit.

8.1 During the Module's assembly, do not twist, bend or add unsuitable external force to the Module.

8.2 Do not use an air gun or blower to clean the Module during the Module's assembly to avoid the contamination of external dust or particles.

8.3 Only operate the Modules in the specified temperature range and a ventilated environment.

8.4 Avoid dust or oil mist during Module assembly.

8.5 Follow the correct power sequence while operating the Module. Module will be damaged otherwise.

8.6 Avoid electrical magnetic interference for higher safety and lesser noise.

8.7 Operate the Module in the specified temperature range as the temperature will affect the response time and brigthness.

8.8 Avoid displaying the fixed pattern (especially the white pattern) for a long period of time as it will cause image sticking.

8.9 Be sure to turn off the power when connecting or disconnecting the circuit.

8.10 Avoid dirt and stain on the surface of modules. As it can easily scratch the surface polarizer.

8.11 Wipe off any moistures before operating the Modules as the dewdrop may demage the Modules.

8.12 Avoid drastic temperature changes, as the condensation or expansion reactions will damage (destroy) the polarizer.

8.13 Avoid exposing the modules under sunglight as the high temperature and humidity will degrade the performance.

8.14 Avoid any harmful materials/ components/ chemiclas on the modules, such as acid or chlorine.

8.15 Connect grounded devices while operating the Module as the static electricity will damage the Module.

8.16 Disassemble or reassemble the Modules are forbidden.

8.17 Avoid direct contact on the rear side as the backlight's high voltage will cause electrical shocks.

8.18 Avoid strong vibration or shock on the Module as they will damage the Module.

8.19 Store the Modules in suitable environment with regular packing.

8.20 Do not press on the surface (either front or rear side) of the Module as it will lead to non-uniformity or other function issues. Be cautious of the broken Module to avoid damages.

8.21 Softly tear the sticking tape of the protective film to prevent bezel out of shape and non-uniform display.

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